

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and can cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Drain the oil and remove the oil pan. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>).
- Remove the oil suction tube and block stiffener plate. Refer to Procedure 001-089 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>).

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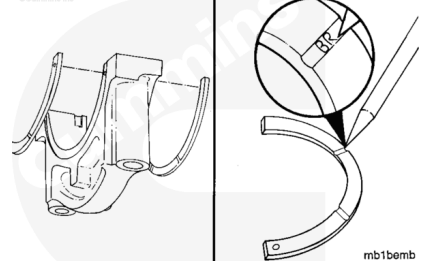
ck800wa

Remove

The number 6 main bearing cap contains two thrust bearings.

Remove and mark them in the notched area for position.

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mb1bemb

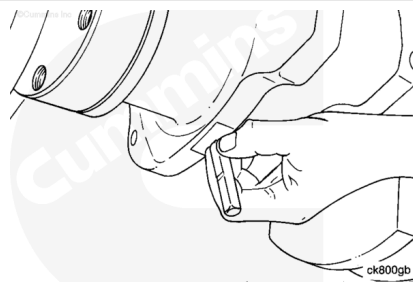
Remove the two upper thrust bearings from the block at the number 6 location. Mark them for position.

If necessary, slide the crankshaft to the front or to the rear to allow the thrust bearings to be removed.



Inspect for Reuse

Thrust bearing sets are available in five sizes. The upper and lower thrust bearings **must** be the same size. The front and rear can be different sizes.

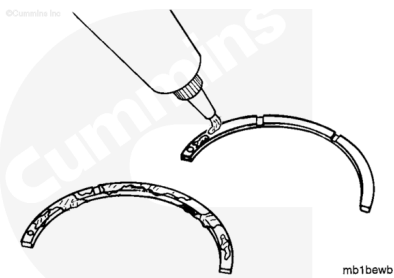


Bearing Set Size	Set Contents
STD	Four (4) STD Bearings
0.25 mm [0.010 in]	Two (2) STD and two (2) 0.25 mm [0.010 in] oversize bearings
0.50 mm [0.020 in]	Four (4) 0.25 mm [0.010 in] oversize bearings
0.75 mm [0.030 in]	Two (2) 0.25 mm [0.010 in] oversize and two (2) 0.50 mm [0.020 in] oversize bearings
1.00 mm [0.040 in]	Four (4) 0.50 mm [0.020 in] oversize bearings

Use the same size as those removed. The crankshaft will be marked on a counterweight adjacent to the thrust location if the thrust flange has been machined for oversize thrust bearings.

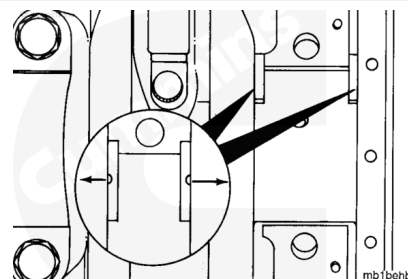
Install

Use clean engine oil to lubricate the thrust bearings.



⚠ CAUTION ⚠

The grooves in the thrust bearings must point toward the crankshaft. Engine damage can occur if the thrust bearing is not orientated correctly.

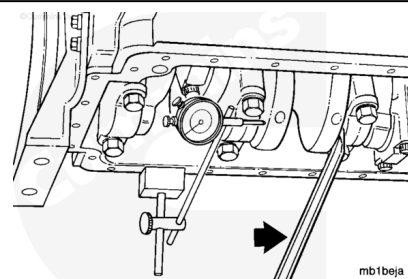


Install the bearings in the number 6 location.

Move the crankshaft to the front or to the rear so the second thrust bearing can be installed.

The upper and lower thrust bearings on the same side of the main cap **must** be the same thickness.

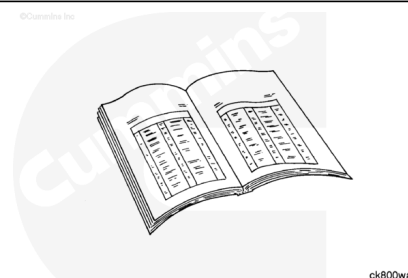
Install the main caps and measure the crankshaft end clearance. Refer to Procedure 001-006 in Section 1.
(/qs3/pubsys2/xml/en/procedures/89/89-001-006.html)



Finishing Steps

⚠ CAUTION ⚠

Do not pry or bend the suction tube to get the support bracket capscrew holes to align. A bent suction tube can cause improper oil flow and cause engine damage.



- Install the block stiffener plate and suction tube. Refer to Procedure 001-089 in Section 1
(/qs3/pubsys2/xml/en/procedures/89/89-001-089.html).

- Install the oil pan and related components. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>).
- Fill the engine with clean lubricating oil. Use the following procedure for engine oil capacity specifications. Refer to Procedure 007-037 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>).
- Operate the engine to 70°C [160°F] coolant temperature and check for leaks.